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Evolutionary history of quadrupedal walking gaits shows mammalian release from locomotor constraint

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Vertebrates employ an impressive range of strategies for coordinating their limb movements while walking. Although this gait variation has been quantified and hypotheses for its origins tested in select tetrapod lineages, a comprehensive understanding of gait evolution in a macro-evolutionary context is currently lacking. We used freely available internet videos to nearly double the number of species with quantitative gait data, and used phylogenetic comparative methods to test key hypotheses about symmetrical gait origin and evolution. We find strong support for an ancestral lateral-sequence diagonal-couplet gait in quadrupedal gnathostomes, and this mode is remarkably conserved throughout tetrapod phylogeny. Evolutionary rate analyses show that mammals overcame this ancestral constraint, resulting in a greater range of phase values than any other tetrapod lineage. Diagonal-sequence diagonal-couplet gaits are significantly associated with arboreality in mammals, though this relationship is not recovered for other tetrapod lineages. Notably, the lateral-sequence lateral-couplet gait, unique to mammals among extant tetrapods, is not associated with any traditional explanations. The complex drivers of gait diversification in mammals remain unclear, but our analyses suggest that their success was due, in part, to release from a locomotor constraint that has probably persisted in other extant tetrapod lineages for over 375 Myr.

1. Introduction

Since the time of the ancient Greek philosophers, biologists have been intrigued by the diversity of limb movements and coordination patterns used by quadrupedal tetrapods [1–6]. The true magnitude of this variation only became evident when Hildebrand [3–5] devised a means to graphically represent symmetrical quadrupedal gait sequences (i.e. where ‘the footfalls of a pair of feet (fore or hind) are evenly spaced in time’ [7], p. 701) using the combination of duty factor (a relative measure of support provided by each limb) and phase (a quantitative measure of lag between limbs). An advantage of these ‘Hildebrand plots’ is that, when many species are graphed together, patterns emerge that appear to reflect differences in ecology [8–11], locomotor modes [12,13] and anatomy [2,10,14].

Walking gaits, defined by a duty factor above 50%, are divided into four categories based on phase (figure 1). Diagonal-sequence diagonal-couplet gaits (DSDC; phase values between 50 and 75), which minimize unilateral limb support and potentially result in increased locomotor costs, are often observed within primates and other highly arboreal taxa [2,8,12,13,15,16]. Lateral-sequence lateral-couplet gaits (LSLC; phase values between 0 and 25), which have been proposed to limit the likelihood of interlimb interference [5,14,17], are more common in terrestrial tetrapods with relatively long limbs. Between these

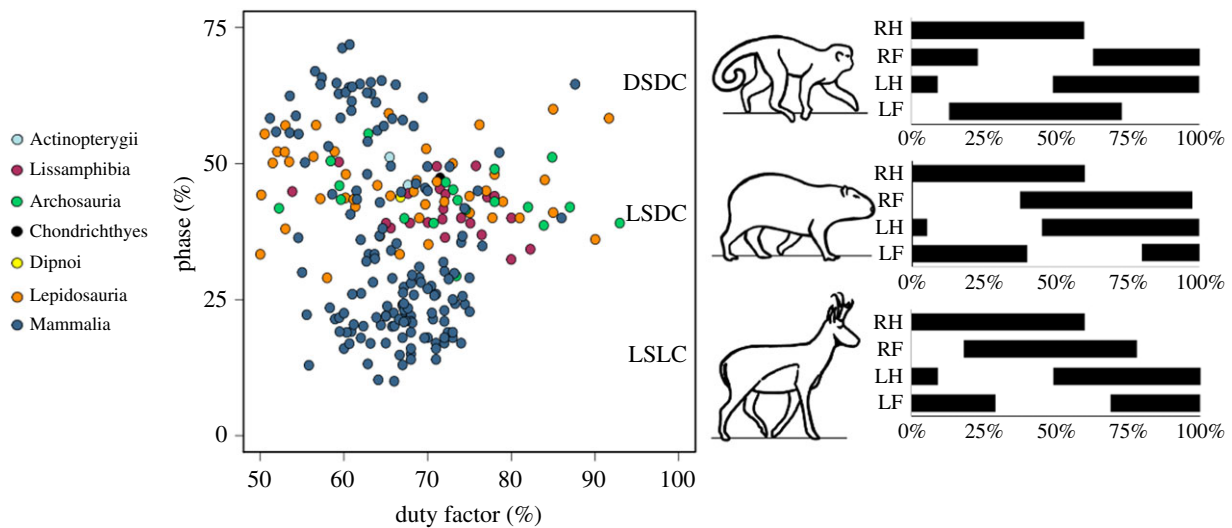


Figure 1. ‘Hildebrand plot’ displaying phase against hindlimb duty factor collected during symmetrical quadrupedal walking (duty factor greater than 50%). Silhouettes on the right side show illustrations of walking gaits relative to phase values (pronghorn antelope as an example of a lateral-sequence lateral-couplet (LSLC) gait, capybara representing lateral-sequence diagonal-couplet (LSDC) gait, and spider monkey showing a diagonal-sequence diagonal-couplet (DSDC) gait). To the right of each silhouette is a corresponding gait diagram demonstrating the proportion and duration of time in which each limb makes contact with the ground, indicated by the horizontal black bars. All gait diagrams are scaled to 60% duty factor. (Online version in colour.)

two extremes falls the lateral-sequence diagonal-couplet (LSDC; phase values between 25 and 50) gait, a gait sequence where contralateral fore- and hindlimbs contact the substrate more-or-less synchronously [3–5,17–19]. LSDC gaits have been proposed to be the primitive condition for tetrapods, though this assumption is based largely on observations from individual species rather than any formal evolutionary analyses [10,20–22]. The LSDC gait is considered to be highly stable, minimizing the possibility of the animal toppling over by maximizing the proportion of the stride in which the centre of mass is positioned over the triangle of support [3–5,14,17–19]. A fourth gait, diagonal-sequence lateral-couplet (phase values between 75 and 100), is theoretically possible but biomechanically unlikely and is rarely observed in nature [17].

While the rules and trends governing which species will adopt which gait within the Hildebrand framework are considered well established, there remain critical pitfalls in the way such notions have been generated. First, taxonomic sampling, with limited exceptions (e.g. [6,10,23]), has been largely mammal-centric [2,3,5,17]. This sampling bias raises the question of whether observed correlations between interlimb coordination pattern, morphology and locomotor behaviour are indicative of quadrupedal gnathostomes in general, tetrapods, amniotes or just mammals. Second, comparative studies have not used any sort of phylogenetic framework within their analyses [24,25]. Therefore, hypotheses about gait patterns specific to arboreal or long-legged taxa may be more influenced by phylogenetic relatedness of the examined taxa, rather than functional necessity. Finally, by not examining gait sequences in an evolutionary context we make the assumption that the observed variation in limb coordination patterns arose based completely on adaptive necessity. While many studies have focused on the influence of substrate use and stability concerns as the primary drivers of phase evolution, a recent numerical model [6] found that mechanical work done by the limbs during symmetrical walking gaits depends simultaneously on phase and duty factor, leading to a predicted positive correlation between these two

factors if selection acts to minimize work demands. Empirical data from a small sample of tetrapods are consistent with this model, albeit with an offset relationship for primates, but no formal evolutionary tests for this hypothesis have yet been conducted. Furthermore, recent work suggests that the neuromuscular origins of quadrupedal symmetrical gaits may predate the origins of tetrapods [22,26,27], raising the possibility that neuromuscular constraints might limit certain lineages from exploring novel gait sequences. Taken together, our current knowledge about the phylogenetic and functional patterns that drove gait evolution in quadrupedal animals is insufficient.

Here, we use crowd-sourced data and phylogenetic comparative methods to provide the first explicit macroevolutionary tests of hypotheses regarding the evolution of quadrupedal walking gaits in gnathostomes, using the phylogenetically broadest comparative gait data set published to date. First, we ask whether the phylogenetic distribution of gaits provides support for an ancestral LSDC gait in gnathostomes and tetrapods. Next, we assess whether the apparent diversity of gaits used by mammals is best explained by elevated rates of gait evolution or release from an evolutionary constraint. Finally, we test whether ecological and functional hypotheses for the evolution of DSDC and LSLC gaits are supported using a broad comparative sample of mammalian taxa.

2. Material and methods

(a) Data collection

We performed an extensive literature review to compile phase and duty factor data from previously published sources (electronic supplementary material, table 1). These were supplemented with unpublished data from MCG, resulting in a dataset spanning 127 species. To expand sampling and generate a more comprehensive comparative dataset for macroevolutionary analysis, we collected quantitative gait data for 117 previously unstudied species using freely available internet videos [6]. Videos were

chosen based on the ability to observe a full gait sequence and the clarity with which footfalls could be observed. We only analysed videos in which animals appeared to be moving at a steady state (i.e. not accelerating or decelerating). We preferred videos in which animals were walking on a flat surface, but some arboreal species were observed in their natural habitat. The use of internet videos allowed us to minimally double our sampling for Actinopterygii, Archelosauria, Lepidosauria and Lissamphibia, as well as adding 61 mammals to our existing sample of 91 species. When several videos were used or multiple strides were observed for a single species, we used the average phase and duty factor over all strides [6]. For calculations of duty factor and phase, as well as discussion of the limitations of our approach and sampling scheme, we refer the reader to the electronic supplementary material, Methods.

Body mass (g) estimates for mammals were taken from the literature (electronic supplementary material, table 2) when available or collected from the PanTHERIA database [28]. Hind limb lengths (combined length of femur, tibia and third metatarsal) were taken from the literature and supplemented with measurements from skeletal specimens primarily at the Field Museum of Natural History (FMNH, Chicago, IL, USA) (electronic supplementary material, table 2). Relative limb length was calculated as raw limb length divided by the cube root of body mass (see the electronic supplementary material, Methods for a discussion on limitations of this approach).

(b) Phylogeny

We downloaded a time-scaled tree for all taxa in our comparative dataset from www.timetree.org [29]. Three taxa in our sample were not represented in timetree and so we downloaded trees with their closest available relatives and substituted taxon names as appropriate: *Ginglymostoma cirratum* (changed to *Hemiscyllium ocellatum*), *Lepidosiren paradoxa* (changed to *Protopterus annectens*), *Hemimyzon taitungensis* (changed to *Cryptotora thamicola*). Zero-length branches, which can cause problems in comparative analyses, were arbitrarily replaced with lengths of 0.01 Myr, with the same amount of time deducted from the parent branch to maintain ultrametricity. We visualized the phylogenetic distribution of phase by mapping it as a continuous trait on our tree using the `contMap()` function from the R [30] package `phytools` [31], assuming a Brownian motion model of evolution. Phase is a proportional variable and so we applied an arcsine square root transformation prior to subsequent statistical analyses with back-transformations applied when interpreting estimates.

(c) Ancestral gait estimation

We tested the hypothesis that LSDC gaits represent the primitive gnathostome and tetrapod condition by estimating ancestral phase values using the `fastAnc()` function in the `phytools` library. Ancestral state estimation for continuous traits typically assumes a Brownian motion model in which character evolution occurs at a constant rate over time and across phylogeny. When this assumption is violated, as it frequently is, fossil data can improve the accuracy and precision of ancestral state inference [32,33]. However, fossil data are not available for behavioural traits, except in rare circumstances, and phase cannot currently be calculated for any extinct taxa. Accounting for heterogeneity in evolutionary rates has also been shown to improve the accuracy of ancestral state estimation [34,35] by reducing the influence of branches along which large amounts of change have occurred while maximizing the signal from branches along which traits have diverged less from their ancestral values, either due to slow rates or constraints [36]. Thus, we also estimated ancestral states using a phylogeny in which

branch lengths were transformed according to the posterior mean estimates of their rates of phase evolution (see below).

(d) Tempo and mode in phase evolution

The diversity of gaits used by extant mammals in comparison with extant non-mammalian quadrupedal gnathostomes (figure 1) suggests that the rate of phase evolution may have accelerated in crown group Mammalia. We tested this hypothesis by estimating per-branch rates and inferring the presence of rate shifts under a Brownian motion model of evolution using a reversible-jump Markov chain Monte Carlo (MCMC) method implemented in BAMB v.2.5.0 (Bayesian analysis of macroevolutionary mixtures [37]). We used the `setBAMMpriors()` function in the `BAMMtools` package [38] to determine priors and ran the MCMC for 10^6 generations, sampling every 10^5 . After discarding the first 20% of samples as burn-in and checking for large effective sample sizes (>200) using the `coda` library [39], we summarized the credible set of shift configurations to determine the number and location of shifts, as well as the marginal probability of shifts occurring along each branch of the phylogeny. To account for variable rates of phase evolution in ancestral state estimation, we used the `BAMMtools` function `getMeanBranchLengthTree()` to generate a tree annotated with mean per branch rates.

Low trait variance can equally result from slow evolutionary rates or from evolutionary constraints [40]. However, constraints on trait diversification should leave a strong signature in comparative data, in the form of reduced phylogenetic signal, as traits fail to diffuse away from their starting point and close relatives are no more similar than distant ones [41]. To test whether mammalian phase diversity is consistent with release from an ancestral constraint, we computed phylogenetic signal for phase in mammals and non-mammalian gnathostomes separately, measured as Blomberg's K [42], using the `phylosig()` function in `phytools`, with 1000 randomizations used to determine whether the observed value differed significantly from random.

(e) Ecological and morphological correlates of phase

We used phylogenetic generalized least squares (PGLS) analysis to test whether LSLC gaits are better explained by relative limb length [3,4,14] or the need to minimize mechanical work done by the limbs, as predicted by Usherwood & Self Davies [6]. For completeness, we also assessed models in which body mass and absolute limb length were treated as the independent variable. PGLS analyses were run using the `glS()` function from the package `nlme`. To appropriately account for phylogenetic autocorrelation in the residual error term we co-estimated Pagel's λ using the `corPagel` structure in `ape` [43].

The strong connection between DSDC gaits and arboreality in mammals has led to a general perception that substrate preference may influence gait [2,4,5,8,11,13,15,16,18,19]. To test this association more robustly in a more comparative sample of quadrupedal tetrapods, we used phylogenetic ANOVAs [44] to determine whether basic differences in habitat (terrestrial versus arboreal) could explain differences in phase values by comparing the means of the two habitat groups for mammals and non-mammalian gnathostomes. Phylogenetic ANOVAs were performed using the function `phylANOVA()` in the package `phytools`, with 1000 randomizations used to determine significance.

3. Results

(a) Phylogenetic patterns

Mapping of phase onto a time-scaled phylogeny (figure 2) confirms that non-mammalian taxa are conservative in interlimb coordination strategies (variance = 0.0045).

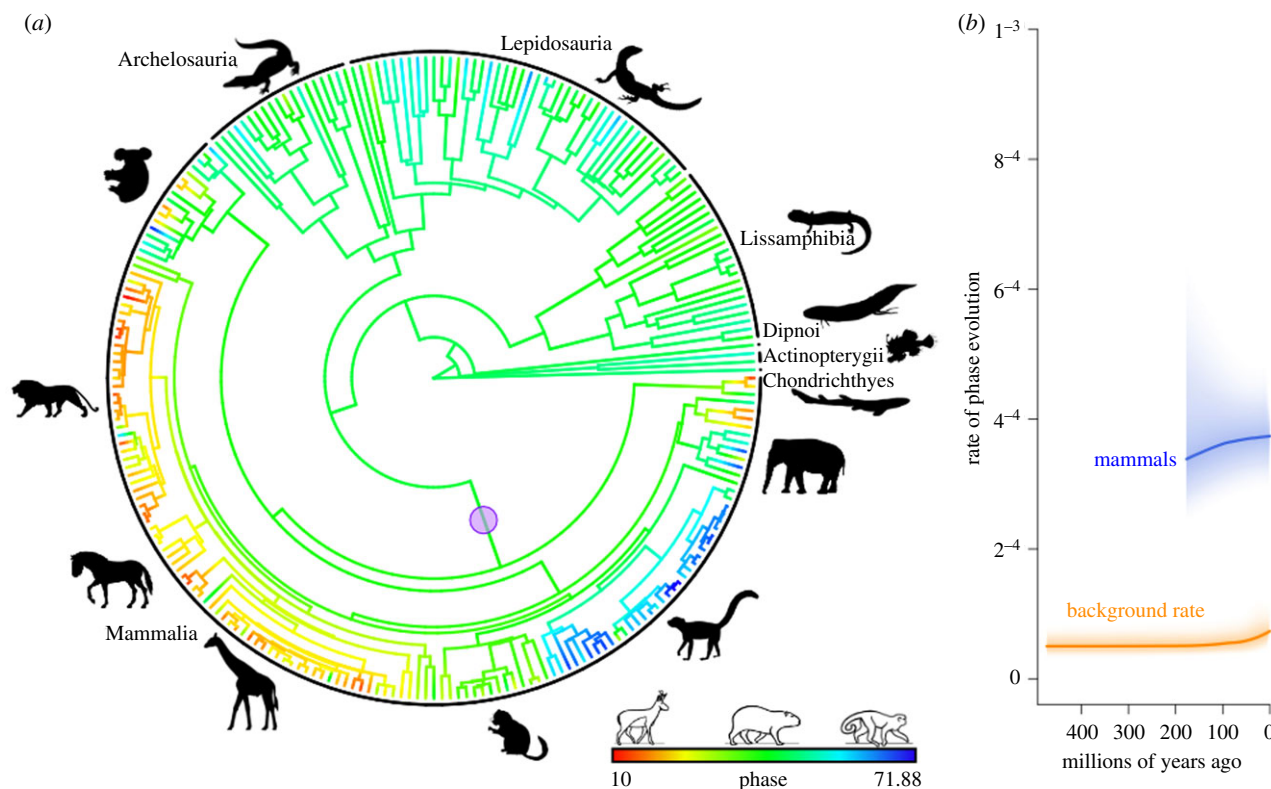


Figure 2. (a) Phylogeny of raw phase values for our dataset of 244 quadrupedal tetrapods and non-tetrapod gnathostome taxa (for a list of taxa included in this study we refer the reader to electronic supplementary material, table S1). The large purple dot indicates the most supported shift in rate of phase evolution, along the branch leading to crown group Mammalia. This figure serves as a visual representation of ancestral state estimations as well, showing the primitive gait sequence to most likely be the LSDC gait. (b) Evolutionary rates of phase in mammalian taxa and all non-mammalian taxa (labelled as 'background rate'). (Online version in colour.)

quadrupedal Non-mammalian quadrupedal gnathostomes primarily adopt LSDC gaits with the occasional venture into DSDC gaits, though not nearly to the degree seen in mammals. By contrast, mammalian taxa are diverse (variance = 0.034) and show a higher proportion of specialized DSDC gaits, with the addition of the novel LSLC gait that is not found outside of this clade.

(b) Ancestral gait estimation

Maximum likelihood estimates of phase for the most recent common ancestors of gnathostomes, tetrapods and amniotes all fall within the range of LSDC gaits (figure 2), albeit with wide confidence intervals that do not preclude the possibility of ancestral LSLC or DSDC gaits (electronic supplementary material, table S3). Using rate-transformed branches derived from BAMM analyses had limited effect on maximum likelihood estimates but resulted in narrower confidence intervals that almost exclusively lie within the range of LSDC values (gnathostomes = 46.76 [41.24–52.32]; tetrapods = 42.44 [37.14–47.83]; amniotes = 42.36 [36.32–48.52]). LSDC gait is also inferred to extend all the way to the mammalian (40.69 [30.29–51.52]) and therian (41.46 [28.87–54.65]) common ancestors, although the possibility of DSDC gaits at these nodes cannot be completely excluded.

(c) Tempo and mode in phase evolution

Five distinct shift configurations were present in the 95% credible set of rate shifts. The most frequently sampled configuration (posterior probability = 0.31), contained a single shift while all others (posterior probabilities = 0.31–0.05) contained two shifts. Background rates of phase evolution are

slow in gnathostomes and both shifts, where present, are to faster rates. The first shift is present in nearly all sampled configurations (marginal probability > 0.99) and represents a roughly fivefold increase in the rate of phase evolution in mammals (figure 2b). This shift is also present in the maximum a posteriori (MAP) configuration (electronic supplementary material, figure S2) and in model-averaged per-branch rates (electronic supplementary material, figure S3). The second shift is not present in the MAP configuration but, where present, results in a moderate to very rapid increase in rate within the Chamaeleonidae or the clade Acrodonta (Chamaeleonidae + Agamidae), though the exact location is uncertain (electronic supplementary material, figures S1–S3).

Phylogenetic signal in phase is low but significantly different from random expectation across gnathostomes (Blomberg's $K = 0.34$, $p < 0.001$). Examining mammals and non-mammalian gnathostomes separately, we found high phylogenetic signal for mammals ($K = 0.76$, $p = 0.01$), consistent with the hypothesis of fast, unconstrained evolution. Non-mammalian gnathostomes exhibit low phylogenetic signal that cannot be distinguished from random ($K = 0.2$, $p = 0.521$). Combined with their low inferred evolutionary rates, this pattern is consistent with an ancestral constraint on phase evolution.

(d) Ecological and morphological correlates of gait

We found support for a negative relationship between phase and body mass (figure 3a; PGLS slope = -0.04 , $p < 0.001$, Pagel's $\lambda = 0.97$) and between phase and absolute limb length (figure 3b; PGLS slope = -0.13 , $p < 0.01$, Pagel's $\lambda = 1$) in our sample of mammalian taxa. However, contrary to Hildebrand's hypothesis, we found a positive but insignificant relationship between phase and limb length after

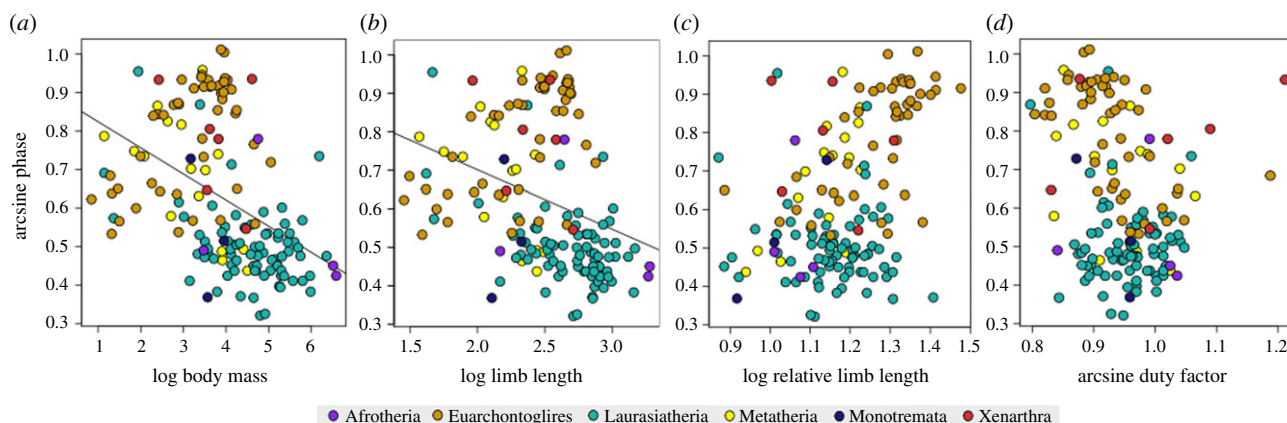


Figure 3. Results from the PGLS analysis. Plots show the relationship between arcsine-transformed phase and (a) log body mass, (b) log limb length, (c) log relative limb length and (d) arcsine-transformed duty factor. (Online version in colour.)

adjusting for body mass (figure 3c; PGLS slope = 0.13, $p = 0.19$, Pagel's $\lambda = 0.97$). We also found no support for a positive relationship between phase and duty factor, as predicted by the Usherwood & Self Davies [6] model (figure 3d; PGLS slope = -0.13, $p = 0.30$, Pagel's $\lambda = 0.97$). Using phylogenetic analyses of covariance, as implemented in *evomap* package [45], we corroborated that the lack of support for these hypothesized relationships cannot be explained due to distinct functional scaling relationships in the long-limbed primates, relative to other mammals (see electronic supplementary material, Methods for details).

Consistent with inference derived from more phylogenetically limited studies, we found significant support for higher phase in mammals that use arboreal substrates ($n = 49$, mean = 50.06, s.d. = 15.83) than those that occupy terrestrial habitats ($n = 105$, mean = 28.05, s.d. = 13.08; phylogenetic ANOVA, $F = 81.3$, $p < 0.001$; figure 4a). However, the relationship between phase and habitat does not hold outside of mammals; arboreal non-mammalian tetrapods have lower mean phase values (mean = 38.9, s.d. = 5.12) than terrestrial taxa (mean = 45.01, s.d. = 6.79), though this difference is not significant (phylogenetic ANOVA, $F = 7.49$, $p = 0.06$) (figure 4b).

4. Discussion

(a) Macroevolutionary patterns in quadrupedal gaits

Our ancestral state estimates provide the first quantitative phylogenetic support for the LSDC gait as the primitive condition in tetrapods. It is particularly notable that, despite the well-known tendency for ancestral states to have large associated errors [46], the 95% confidence intervals for key nodes (e.g. gnathostomes, tetrapods and amniotes) exclude the possibility of alternate gaits after accounting for variation in evolutionary rates. Fossil evidence for an aquatic origin of gait-like patterns of limb movement [20,21,47] has led some authors to propose extant elasmobranch and actinopterygian models for understanding early quadrupedal tetrapod locomotion. In particular, the epaulatte shark (*Hemiscyllium ocellatum*) can maneuver well in shallow, rocky environments and uses its long tail as an additional support structure, a configuration that has been proposed for the stem tetrapod *Acanthostega* [21]. Additionally, the walking cavefish (*Cryptoptora thamicola*) exhibits a gait-like pattern of movement with similarities to a submerged Devonian trackway [48]. It has

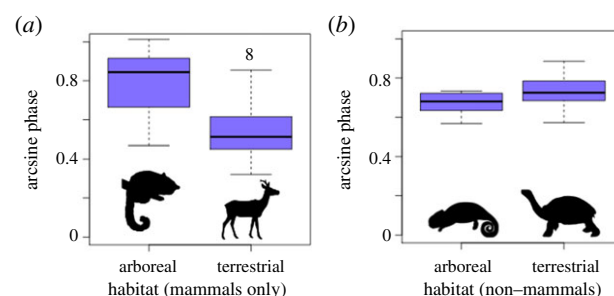


Figure 4. Comparative box plots demonstrating the difference between arcsine-transformed phase values in arboreal and terrestrial taxa for (a) mammals and (b) non-mammals. (Online version in colour.)

been argued, based on *in silico* assessment of 3D limb mobility in the Late Devonian *Ichthyostega*, that at least some early tetrapods would not have been capable of using walking gaits and that a mudskipper-like 'crutching' mode of locomotion was more likely [49]. Our results are not in conflict with such an interpretation of key fossil taxa. Rather, they suggest that instead of representing a novelty associated with the transition to land, the propensity to use LSDC walking gaits is hard-wired in the vertebrate neuromuscular repertoire and predates the emergence of tetrapods. Such an interpretation is in accordance with developmental [26], experimental [20,21,27,50] and paleontological [51] evidence.

Our phylogenetic analyses further suggest that terrestrial vertebrates were likely locked in to the use of LSDC gaits for over 300 Myr, until mammals were able to break this ancestral constraint. This shift occurred somewhere along the branch leading to crown Mammalia, though without fossil data (see below) it is difficult to tell where in time new footfall sequences first appeared. Ancestral state estimation suggests that the common ancestor of monotreme and therian mammals retained a primitive LSDC gait. Flexibility of the axial skeleton and pelvis in therian mammals, as well as the evolution of the therian shoulder girdle, probably increased the ability to move through all three degrees of freedom, which may have further allowed for increased coordination when using symmetrical gaits in this clade [52]. However, the 135 Myr stem lineage leading to the mammalian crown is well represented in the fossil record and it is possible that locomotor experimentation in non-mammalian synapsids [53] may have yielded earlier changes in footfall

sequence. Recent work [54] has shown that functional regionalization of the axial skeleton, a key characteristic of modern mammals, post-dated morphological regionalization and occurred relatively late in synapsid history, probably by the Late Triassic with the appearance of tritylodont cynodonts. The Jurassic mammalian adaptive radiation [55] resulted in many similar ecomorphotypes to those observed in mammals today [56]. It was at this time that mammaliaformes also underwent increasing encephalization and neocortical expansion [57]. The neocortex is largely made up of a somatosensory field that receives information from mechanoreceptors located throughout the skin and musculature [58]. Improved monitoring from these peripheral mechanoreceptors has been shown to have direct consequences on the variability of limb kinematics and kinetics during quadrupedal locomotion [59,60]. An additional neocortical motor map is associated with skilled movements that depend on fine control of distal musculature [57,58]. The resulting rise in sensory input and enhanced neuromuscular coordination may have been exapted and over time provided mammals with more precise neuromuscular control over locomotion.

Although sampling in our study is numerically and phylogenetically broader than any previous analysis of walking gait evolution, potential biases remain. Despite efforts to diversify our sample, we retain a mammal-dominated dataset. It is therefore possible that we have failed to identify independent origins of DSDC and LSLC gaits or that we have missed bursts of gait diversification in clades other than mammals. We recovered some support for an additional pulse of gait diversification with acrodontan squamates, though the exact location of this shift remains unclear. It is notable that this clade includes the Chamaeleonidae, which exhibit some of the most unusual locomotory adaptations found within Squamata [61,62]. Further quantification of walking gaits in acrodontan squamates may help to clarify the nature of this macroevolutionary event. It would also be of interest to quantify phase diversity in other model macroevolutionary systems, such as Antillean anoles [63,64], to understand whether replicated patterns of niche use and morphological evolution result in concordant convergence in walking gaits.

Our sample further constitutes only a small portion of tetrapods that have ever lived. Our conclusions might, therefore, overlook any significant gait variation in morphologically diverse extinct clades not represented here. For example, the living archosaurs represented in our dataset (birds and crocodilians) move very differently from each other but do not deviate from an LSLC gait when walking, though crocodilians can perform a diversity of running gaits [65]. However, Triassic archosauromorphs achieved a great variety of postcranial morphologies that suggest unique locomotor modes relative to other living and extinct reptiles [66–70]. Archosaur diversity is largely attributed to changes in limb posture and stance [66–70], but these changes alone are unlikely to lead to changes in footfall sequence. In the case of mammals, monotremes have a sprawling posture relative to therian mammals but the echidnas (*Zaglossus bruiinji* and *Tachyglossus aculeatus*) use an LSLC gait, indicating that gait sequences are not limited by parasagittal versus sprawling limb postures. Trackways represent an alternative possible source of information for assessing gait diversity in extinct animals, but their utility is limited for two reasons; the track-maker is often a point of contention [67,71] and, though gait

can be speculated on using extant comparisons [72], it is nearly impossible to quantitatively assign phase and duty factor. Though we think it is unlikely that any extinct clades achieved the degree of gait diversity exhibited by extant mammals, our results remain to be corroborated by future tests that are able to leverage the rich tetrapod fossil record to its fullest extent.

(b) Gait, morphology and ecology

There has been considerable debate concerning whether diagonal-sequence diagonal-couplet gaits are adapted for arboreal locomotion, or if they are simply a primate neuromuscular synapomorphy that has misled adaptationist thinking. Our broad comparative study reveals that DSDC gaits are indeed an arboreal adaptation in mammals, having evolved convergently in such divergent taxa as primates, carnivorans, xenarthrans, and diprotodont and didelphimorph marsupials [13,15–17]. Studies of locomotion in arboreal mammals have emphasized the benefits of DSDC gaits by showing minimized periods of unilateral support by ipsilateral limbs and allowing animals with grasping extremities to take hold of the support with the opposite hindfoot if the leading forefoot comes down on a support that fails [13,15]. However, some arboreal mammals in our sample, such as squirrels (*Sciurus* sp.) and tree shrews (*Tupaia* sp.) use LSLC gaits that are more typical of generalized terrestrial locomotion and recent studies have highlighted that many small-bodied (less than 200 g) mammals do not consistently use symmetric gaits on arboreal substrates [73,74]. Together, these findings suggest that DSDC gaits are only beneficial when body size is larger than support size, or for taxa that use the visual system to guide forelimb placement during walking [23]. Although a few squamates within our sample use DSDC gaits (e.g. *Leiocephalus*, *Xenosaurus* and *Eulamprus*), all are terrestrial, and arboreal species such as *Anolis*, *Gekko* and the most specialized arboreal squamate clade, chameleons [62], use LSLC gaits. This finding supports the presence of a neuromuscular constraint in non-mammalian quadrupedal gnathostomes. There is a pressing need for more quantitative phase data from non-mammalian tetrapods to fully untangle these hypotheses.

While the mammalian gait literature has largely focused on DSDC gaits and its association with arboreality, our quantitative comparative data show that the true mammalian innovation is the LSLC gait. However, we find no support for either of the two main functional hypotheses for the adoption of LSLC gaits. Although LSLC gaits are common within large, long-legged terrestrial mammalian clades (Artiodactyla, Perissodactyla and Carnivora; figure 2), we found no significant effect of relative limb length on phase [3,4,14]. Hildebrand's relative limb length hypothesis was derived primarily from observations of gait variation in a sample of domestic dogs classified into relatively long-limbed and relatively short-limbed groups [14], thus it is perhaps unsurprising that such patterns fail to hold across mammals more generally. Usherwood & Self Davies [6] proposed an alternative explanation for LSLC gaits, namely that limb phase is not influenced by stability or interlimb interference, but is directly driven by mechanical work requirements for a given duty factor. Again, our broad comparative sample lends no support to this hypothesis, even after accounting for the distinctive gaits employed by

arboreal primates [6,23]. Taken as a whole we cannot, at present, confidently advance a singular explanatory variable to predict the range of limb phase patterns observed in mammals. The significant negative relationships recovered for phase with body mass and absolute limb length (figure 3*a,b*) suggest a simple scaling explanation for LSLC gait evolution, but the generally high scatter about the PGLS slopes (note that the concept of R^2 is of limited use in generalized least squares) warrant caution in drawing firm conclusions. Nevertheless, our data refute the almost axiomatic association between relative limb length and phase and suggest that the mechanism underlying mammalian gait variation remains to be identified.

(c) Conclusion

Our study suggests that the first tetrapods inherited a propensity for LSDC gaits from their aquatic ancestors and that this gait persisted for over 375 Ma as the dominant strategy of interlimb coordination in quadrupedal tetrapods, until the present day. As mammals proliferated, so did their ability to evolve novel footfall patterns, resulting in a remarkable diversity of gaits traditionally thought to be unstable. The emergence of these new locomotor strategies likely reflected changes in habitat, with arboreal mammals becoming uniquely dependent on the DSDC gait for stability in a complex environment. An arboreal/terrestrial dichotomy is not seen in non-mammalian species, indicating that

neuromuscular coordination may influence the ability to adopt such an advanced mode of interlimb coordination. The LSLC gait, a mammalian innovation, is not observed in any other tetrapod lineage and thus should be a focus of future research. Considering the evolutionary history of quadrupedal gait sequences has provided a new framework for investigating patterns of quadrupedal locomotor strategies and has highlighted the degree to which mammals have achieved locomotor diversity in contrast to other extant quadrupedal animals.

Data accessibility. Additional methods and results supporting this article have been uploaded as part of the online electronic supplementary material. All datasets and scripts used in this study are available from Dryad Digital Repository: <https://doi.org/10.5061/dryad.z08kprd5> [75].

Authors' contributions. A.N.W.: conceptualization, data curation, formal analysis, investigation, methodology, project administration, visualization, writing-original draft, writing-review and editing; G.J.S.: formal analysis, investigation, methodology, writing-original draft, writing-review and editing; M.C.G.: conceptualization, data curation, investigation, methodology, project administration, supervision, writing-original draft, writing-review and editing.

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